ABERDARE TECHNICAL COLLEGE

MODULE 3,TERM 3

UNIT:NETWORK DESIGN AND MANAGEMENT

CONTINUOUS ASSESSMENT TEST-CAT2

UNIT CODE: 0612 451 07A

ELEMENT:INSTALL COMPUTER NETWORK

Total marks: 50

Shirlene Karimi

962.

Time 1hour 30 minutes

INSTRUCTIONS TO CANDIDATES

1. *This paper has **fifteen questions**. Answer **ALL** questions.*
2. *Write your name, registration number, class and date on the answer booklet.*
3. *The marks for each question or part are shown in brackets. **Each mark matches one point, item or command asked for.** For example, "state four" carries 4 marks.*
4. *In calculation questions, **write the formula first, then substitute, then give the answer**. An answer without a formula or working scores at most half of the marks for that part.*
5. *IP addresses and subnet masks must be written in **dotted-decimal notation** unless stated otherwise.*
6. *Router and switch commands must be written in **Cisco IOS syntax**.*
7. *Use the marks to decide how much to write and how long to spend (about 2 minutes per mark).*
8. *Calculators are allowed. Mobile phones are **not** allowed in the examination room.*
9. *Write neatly. Untidy work may lose marks.*

CamScanner

QUESTIONS

1. List the **four** layers of the TCP/IP model. (4 marks)

2. State **two** functions of the Network layer of the OSI model. (2 marks)

3. Define each of the following terms:
(i) Classful IP addressing (1 mark)
(ii) Classless Inter-Domain Routing (CIDR) (1 mark)

4. State **two** differences between static and dynamic IP addressing. (2 marks)

5. State **two** differences between IPv4 and IPv6. (2 marks)

6. (a) State the default subnet mask of each of the following address classes: Class A, Class B and Class C. (3 marks)
(b) Identify the class of the address **192.168.5.7**. (1 mark)

7. (a) Write the formula used to calculate each of the following in CIDR:
(i) Number of host bits (1 mark)
(ii) Total number of host addresses in a network (1 mark)
(iii) Number of usable host addresses in a network (1 mark)

(b) A network has the address **192.168.10.0/26**. Showing your working, determine:
(i) The number of host bits (1 mark)
(ii) The total number of host addresses (1 mark)
(iii) The number of usable host addresses (1 mark)

8. Convert each of the following CIDR prefix lengths to a subnet mask in dotted-decimal notation:
(a) /24 (1 mark)
(b) /26 (1 mark)
(c) /28 (1 mark)
(d) /30 (1 mark)

CamScanner

9. Convert each of the following subnet masks to a CIDR prefix length:

(a) 255.255.255.0 (1 mark)

(b) 255.255.255.224 (1 mark)

(c) 255.255.255.252 (1 mark)

10. A network must support **at least 50 usable host addresses**. Showing your working, determine:

(a) The number of host bits needed. (1 mark)

(b) The CIDR prefix length. (1 mark)

(c) The subnet mask in dotted-decimal notation. (1 mark)

(d) The number of usable host addresses the network provides. (1 mark)

11. State **two** advantages of CIDR over classful IP addressing. (2 marks)

12. (a) Name the protocol used to assign IP addresses to devices dynamically. (1 mark)

(b) State **one** reason why a router interface should use a static IP address. (1 mark)

13. Write the Cisco IOS command for each of the following tasks on a router:

(a) Set the hostname to **R1**. (1 mark)

(b) Set an encrypted privileged EXEC password **kcse2026**. (1 mark)

(c) Enter configuration mode for interface **GigabitEthernet0/0**. (1 mark)

(d) Assign the IP address **192.168.10.1** with the mask for a /26 network. (1 mark)

(e) Activate the interface. (1 mark)

(f) Save the running configuration to NVRAM. (1 mark)

14. Write the **three** commands used to configure the management interface (VLAN 1) of a switch with the IP address **192.168.10.2** and the mask for a /26 network, and make the interface active. (3 marks)

15. Router R1 connects to the network **10.0.0.0/8** and to the LAN **192.168.10.0/24**. Write the **four** commands needed to configure **RIP version 2** on R1 so that both networks are advertised. (4 marks)

CamScanner

1. Network Access Layer
Internet layer.
Transport Layer.
Application Layer.

2. It handles routing and logical addressing.
It determines the best physical path for data to reach its destination across different networks.

3. Classful IP addressing is a method of assigning IP addresses where addresses are divided into five classes.
Classless Inter-Domain Routing is a method allocating IP addresses that allows networks to use variable-length subnet masks (VLSM) instead of fixed IP address classless.

4. Static IP Address does not change while Dynamic IP Address changes automatically.
Static IP Address is manually assigned while Dynamic IP Address is assigned by DHCP.

5. IPv4 uses 32 bits and is written as four numbers separated by dots while IPv6 uses 128 bits and provides many more addresses than IPv4.
IPv4 is the original IP addressing system while IPv6 is the newer IP system created to replace IPv4.

6. Subnet masks of:
Class A: 255.0.0.0
Class B: 255.255.0.0
Class C: 255.255.255.0

6b) Class C

7. Formula.
i) Host bits = 32 bits - CIDR number
ii) Total number of host addresses
= 2^(Host bits)
iii) The number of usable host addresses
= 2^(Host bits) - 2

8a) /24 → 255.255.255.0

b) /26 = 255.255.255.192. ✓ 1
c) /25 = 255.255.255.240 ✓ 1
d) /30 = 255.255.255.252. ✓ 1

9. a) ~~250~~ 255.255.255.0 = /24. ✓ 1
255.255.255.224 = /27 ✓ 1
255.255.255.252 = /30 ✓ 1

10. at least 60 usable host addresses
a) $2^6 - 2 = 64$ $64 - 2 = 62$ ✓ 1
$= 62$
b) CIDR Prefix = 32 - hostbits
32 - 6 = 26 = /26 ✓ 1
c) /26 = 255.255.255.192. ✓ 1
d) $2^6 - 2 = 64$
$64 - 2 = 62$. ✓ 1
= 62 usable host addresses.

11. ↑ allows flexible network sizes 1
↓ reduces wastage of IP addresses ✓ 1

12. DHCP → Dynamic Host Configuration Protocol ✓ 1
b) To ensure the router has a fixed ✓ and consistent IP address for reliable communication and routing. 1

13. a) ~~Host~~ hostname R1 ✓ 1
b) enable secret kes2026 ✓ 1
c) interface GigabitEthernet 0/0 ✓ 1
d) ip address 192.168.10.1 ✓
Subnet mask: 255.255.255.192. 1/2 (should be in one line)
e) no shutdown. ✓ 1
f) copy running-config startup-config ✓ 1 05.

14. interface vlan 1 ✓ 1
ip address 192.168.10.2 ✓
255.255.255.192 ✓
no ~~shut down~~ shutdown ✓ 1

03

22

15. router rip ✓ 1
version 2 ✓ 1
network 10.0.0.0 ✓ 1
network 192.168.10.0 ✓ 1.

7b) i) Host bits - CIDR Number.

32 - 26 = 6

= 6 host bits.

ii) 2^n (host bits) = Total number of host addresses.

$2^n(6) = 128$

iii) The number of usable host bits.

2^n (host bits) - 2

128 - 2 = 126

= 126

8½